Independent relative *whose*

Fragment 2 — 2026-05-07

This fragment tests whether Pullum's bridge-linked dominance-chain machinery can be used for an empirically delicate CGEL construction: relative *whose* without an overt following possessum.

1. Phenomenon

Independent relative *whose* is the relative-genitive construction in which *whose* appears without a following noun:

(1) *I was going to visit Lucy, a friend of whose had told us of the accident.*

(2) *Norway, whose standard of living is the envy of the world, has thicker kids than Singapore, whose isn't.*

(3) *The man whose these are hath gotten me with child.*

The construction is rare and pragmatically constrained. The syntactic task for this fragment is narrower: state the tree condition under which a relative clause contains a relative-genitive phrase whose possessum is not overtly realised, and connect that phrase to a vacancy or predicative gap in the relative-clause nucleus.

2. Vocabulary

The definitions below adapt Pullum's handout notation. Immediate dominance is $\lhd$; dominance is $\lhd^*$. The relation Bridge(x, y) is initially limited to Head and Complement links:

(4) $\text{Bridge}(x, y) \equiv_{\text{def}} \text{Head}(x, y) \lor \text{Comp}(x, y)$
y is linked to x by a bridge-preserving function

We add RelWhose(w) for a lexical relative-genitive *whose* node and PossRec(w) for the discourse condition that the missing possessum is recoverable. The second predicate is not syntactic; it marks the point where this construction is known to need information-structural licensing. The tree condition below is stated without PossRec; a separate licensed-use condition adds that bridge explicitly.

3. Specification

Pullum's basic vacancy predicate can be stated as:

(5) $\text{Vacancy}(x) \equiv_{\text{def}} \text{Phrasal}(x) \land \neg(\exists y)[x \lhd y]$
a phrasal node with no children

(6) $\text{NPVacancy}(x) \equiv_{\text{def}} \text{Vacancy}(x) \land \text{NP}(x)$
an NP vacancy

The relevant chain definitions are:

(7) $\text{DomOrdered}(X) \equiv_{\text{def}} (\forall x)(\forall y)[(X(x) \land X(y)) \Rightarrow (x \lhd^* y \lor y \lhd^* x)]$.
all nodes in X are ordered by dominance

(8) $\text{DomContinuous}(X) \equiv_{\text{def}} (\forall x)(\forall y)(\forall z)[(X(x) \land X(z) \land x \lhd^* y \land y \lhd^* z) \Rightarrow X(y)]$.
X contains the intervening nodes on its dominance path

(9) $\text{DomChain}(X) \equiv_{\text{def}} \text{DomOrdered}(X) \land \text{DomContinuous}(X)$
a dominance chain

(10) $\text{BridgeLinked}(X) \equiv_{\text{def}} \text{DomChain}(X) \land (\forall x)(\forall y)[(X(x) \land X(y) \land x \lhd y) \Rightarrow \text{Bridge}(x, y)]$.
a dominance chain whose immediate links are bridge links

(11) $\text{Root}(X, r) \equiv_{\text{def}} (\forall x)[X(x) \Leftrightarrow r \lhd^* x]$
X is the set of nodes dominated by root r

(12) $\text{NPBindable}(r) \equiv_{\text{def}} (\exists X)[\text{Root}(X, r) \land \text{BridgeLinked}(X) \land (\exists z)[X(z) \land \text{NPVacancy}(z)]]$.
r roots a bridge-linked constituent containing an NP vacancy

For independent relative *whose*, first state the syntactic link:

(13) $\text{SynWhoseLink}(c, w, r) \equiv_{\text{def}} \text{Clause}(c) \wedge \text{RelWhose}(w) \wedge c \lhd^* w \wedge c \lhd^* r \wedge \text{NPBindable}(r)$.
a clause, a relative-genitive whose, and a bindable nucleus

(14) $\text{SynWhoseCl}(c) \equiv_{\text{def}} (\exists w)(\exists r)[\text{SynWhoseLink}(c, w, r)]$
a syntactic independent-relative-whose configuration

A licensed use adds a discourse bridge:

(15) $\text{LicWhoseCl}(c) \equiv_{\text{def}} (\exists w)(\exists r)[\text{SynWhoseLink}(c, w, r) \wedge \text{PossRec}(w)]$.
a syntactic configuration plus recoverability of the missing possessum

The last definition is not a tree-only predicate. It is an interface claim: $\text{PossRec}(w)$ has to be warranted independently by contrastive parallelism, deictic anchoring, structural anchoring, or some other discourse condition.

4. Worked instance

In *Singapore, whose isn't*, the relative clause contains a relative-genitive *whose* and a nucleus with stranded auxiliary *isn't*. The missing possessum is recoverable from the parallel antecedent *Norway, whose standard of living is ….* In this use, $\text{PossRec}(w)$ is licensed by contrastive parallelism.

The syntactic part then asks whether the nucleus rooted at r contains a bridge-linked NP vacancy. If it does, r satisfies $\text{NPBindable}(r)$ and the relative clause satisfies $\text{SynWhoseCl}(c)$. A separate bridge claim, $\text{PossRec}(w)$, is warranted here by contrastive parallelism. The formula therefore keeps two claims apart: the structural possibility of a relative-genitive dependency, and the pragmatic recoverability of the possessum.

5. Open questions

(a) Some attested cases are predicative-complement cases rather than simple NP-vacancy cases. Those may require a second vacancy type.
(b) Oblique genitive examples such as *a friend of whose* may be locally licensed by the possessum noun inside the relative phrase, rather than by a distant vacancy in the nucleus.
(c) PossRec is a placeholder for discourse licensing, especially contrastive parallelism, deictic anchoring, and structural anchoring. It belongs in LicWhoseCl, not in the syntactic predicate SynWhoseCl.
(d) CGEL page references are available in the separate *whose* project but should be rechecked against the book before citation.